Syllabus (OBE): Autumn 2026

Microprocessor & Assembly Language — Theory and Sessional

ISCED Code: 0714 — CCE-3501 — Course Title: Microprocessor & Assembly Language

Credit Hours: 3

Contact Hours: 3 CH per Week

Prerequisite Course: CCE-2411

Course Assessments

- CIE (Continuous Internal Evaluation): Attendance — 10 Marks
- CIE: Class Test / Assignment / Quizzes — 10 Marks
- CIE: Mid-term — 30 Marks
- SEE (Semester End Examination) — 50 Marks

Course Objective

The objective of this course is to become familiar with the architecture and the instruction set of an Intel microprocessor. Assembly language programming will be studied, as well as the design of various types of digital and analog interfaces. The accompanying lab is designed to provide practical hands-on experience with microprocessor software applications and interfacing techniques.

- To provide the students on developing of assembly level programs parallel providing the basics of processors.
- To provide students with a solid foundation in interfacing external devices to the processor according to the user requirements to create novel products and solutions for real-life problems.
- To assist the students with an academic environment aware of excellence, guidelines, and lifelong learning needed for a successful professional career.

Course Learning Outcomes (CLOs)

- CLO-1: Gain knowledge of assembly-level programming to write and test programs of moderate complexity for a real-time machine. — Bloom's Taxonomy: Cognitive (Understanding); PLO-1
- CLO-2: Apply knowledge to design circuits for various applications using Microcontrollers. — Bloom's Taxonomy: Cognitive (Understanding); PLO-2

Course Contents

Contents	CLOs	Lecture	Tutorial	Practical
Mid Term Examination (30 Marks)				
Introduction to the course, Discussion of Lesson Plan	CLO-1	2	0	0
Basic architectures: Von Neumann and hardwired; evolution of microprocessors; computer generations and classifications; mainframe, Mini, and supercomputers	CLO-1 & 2	2	0	0
Classifications, Mainframe, Mini, Micro and super computers		2	0	0
Microcomputer block diagram, different input/output devices, different memory devices (Primary and Secondary)		2	0	0
Difference between microprocessor, microcomputer, microcontroller		1	0	0

Contents	CLOs	Lecture	Tutorial	Practical
Concept of multiprogramming, multiprocessing, multiuser system, parallel processing	CLO-2	1	0	0
Introduction to Microprocessor 8085, 8086, and 80386 Pin-out diagram and pin architecture, instruction, and data flow	CLO-1 & 2	1	0	0
Microprocessor 8085, 8086 addressing modes, Instruction fetch & execute, machine cycles, instruction cycle, timing diagrams, state transition diagram, instruction groups and instruction set		1	0	0
Semester End Examination (SEE) = 50 Marks — Part-A (20 Marks)				
Basic Assembly language programming: Simple arithmetic program	CLO-1	1	0	0
Assembly Language: Data transfer array, string, bit manipulation, code conversion, delay routine and stack operations		1	0	0
I/O interfacing and memory system design: address design, need for I/O, I/O mapped I/O, memory-mapped I/O	CLO-2	1	0	0
Data transfer schemes: PPI, 8255, its modes, interface, initialization, and programming		1	0	0
Interfacing of key, LED, seven segments		1	0	0
Brief discussion of dot matrix display, ADC 0808		2	0	0
DAC 0809 with PPI 8255, address decoder, and memory system design interface		2	0	0
Stepper motor controller interfacing with 8255		2	0	0
Part-B (30 Marks)				
8251, 8253, 8257, 8279, interface with 8085	CLO-2	2	0	0
Initialization and application-oriented programming. Data transfer schemes		1	0	0
Interrupts: SIM/RIM, PIC 8259 interface		1	0	0
Initialization and programming with 8085, microprocessor system design, 8085 applications such as: programmable logic controller, temperature controller		2	0	0
Review		1	0	0
Total		**18**	**0**	**0**

Textbook

- Barry B. Brey, "The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, and Pentium Pro Processor Architecture, Programming, and Interfacing", Prentice Hall, 5th edition, 2000.

Reference Book

- Ramesh Gaonkar, "Microprocessor Architecture, Programming and Applications with the 8085", Penram International Publishing, 6th edition, 2013.

Course Assessment Pattern (Theory Course)

Bloom's Category	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (SEE) (50)	Total (100)
Remember	-	5	-	5	10
Understand	-	-	5	5	10
Apply	-	5	-	5	10
Analyze	-	5	-	15	20
Evaluation	-	10	5	10	25
Create	-	5	-	10	15
Responding (Affective)	x	x	x	10	10

Note: CIE = Continuous Internal Evaluation, SEE = Semester End Examination.

Remarks

Course teachers may change the magnitude of marks in Bloom's category (both for CIE and SEE), but must keep in mind that the percentage of Higher Order Thinking Skills (HOTS) should be about 60% or more, and all Bloom's categories should be addressed during the semester. If necessary, a course teacher may also use the Cognitive (Knowledge), Affective (Attitude), and Psychomotor (Skills) domains of Bloom's Taxonomy. Question papers for the Semester End Examination (SEE) shall be moderated by the examination committee composed of both internal faculty and an external member.

Delivery Methods & Activities

Lecture, White Board Writing, Questions and Answers, Tutorial, Discussions, PowerPoint Presentation.

Assessment Tools

Class Attendance, Class Test, Quizzes/Assignment on problem solution, Mid-Term & Semester End Examination, Project evaluation & Viva.

ISCED Code: 0714 — CCE-3502 — Course Title: Microprocessor & Assembly Language Sessional

Credit Hours: 1

Contact Hours: 2 CH per Week

Prerequisite Course: CCE-2412

Course Assessments

- CIE (Continuous Internal Evaluation): 40 – 60 Marks
- SEE (Semester End Examination): 60 – 40 Marks

Course Objective

- To enable students on developing of assembly level programs while providing the basics of processors.
- To gain basic knowledge of assembly language and write programs for arithmetic instructions.

Course Learning Outcomes (CLOs)

- CLO-1: Analyze, write, and test assembly language programs of moderate complexity. — Bloom's Taxonomy: Cognitive (Understanding); PLO-1

Course Contents

Contents	CLOs	Lecture	Tutorial	Practical
Introduction to Assembly Language	CLO-1			3
Write an Assembly program for printing 'Hello World'				3
Write an Assembly program to understand the use of registers in an Assembly program				3
Write an Assembly program that reads a number from the keyboard and displays it on the screen				3
Write an Assembly program for arithmetic instructions (Addition, Multiplication, Division)				3
Write an Assembly program for logical instructions ('AND' & 'OR' operations)				3
Total		**0**	**0**	**18**

Textbook

- Lab Manual prepared by course instructor.

Course Assessment Pattern

There are 100 marks for each Sessional course. Out of 100 marks, 40–60 marks are allotted for continuous assessment on lab activities, including 10 marks for attendance (CIE), and 60–40 marks are for the practical exam at the end of the semester, viva, quiz, etc., at the end-of-semester final examination (SEE).